

Equilibrium lesser Green's function of the YSR Josephson contact:
 why $G^< = f (G^A - G^R)$, and what it equals as a bath $\Sigma^<$

This note derives the equilibrium ($V = 0, T = 0$) lesser Green's function used in `currentPhi_eq_Tfull` (the current-phase relation $I(\phi)$ of a current-biased/equilibrium Josephson contact with a classical-spin/YSR impurity, in the 4×4 Nambu $\otimes$ spin basis). It shows that the compact “FDT” form

$$G^< = f f \cdot (G^{R\dagger} - G^R), \quad f f \equiv \theta(-\omega),$$

is *algebraically identical* to the explicit bath construction $G^< = G^R \Sigma^< G^A$ with a specific, clean $\Sigma^<$ — and in particular that the impurity potential, the tunnelling, and any flat “ $2i\Gamma$ ” term must *not* appear in $\Sigma^<$.

1 Model and notation

At real frequency ω define $z = \omega + i\Gamma$ ($\Gamma =$ Dynes broadening). The clean BCS surface Green's function of each lead is

$$g_0(z) = \frac{1}{\zeta \sqrt{\Delta^2 - z^2}} \begin{pmatrix} -z & 0 & 0 & \Delta \\ 0 & -z & -\Delta & 0 \\ 0 & -\Delta & -z & 0 \\ \Delta & 0 & 0 & -z \end{pmatrix}, \quad g_0^A \equiv g_0^\dagger = g_0(z^*), \quad z^* = \omega - i\Gamma. \quad (1)$$

The on-site impurity is Hermitian,

$$V_{\text{imp}} = \begin{pmatrix} \mathbf{J} \cdot \boldsymbol{\sigma} & 0 \\ 0 & -(\mathbf{J} \cdot \boldsymbol{\sigma})^* \end{pmatrix} + K \tau_z \otimes \sigma_0, \quad V_{\text{imp}}^\dagger = V_{\text{imp}}. \quad (2)$$

The impurity-dressed retarded surface GF of lead $\alpha = L, R$ is

$$g_{r\alpha} = (\mathbb{I} - g_0 V_{\text{imp}}^\alpha)^{-1} g_0 \implies g_{r\alpha}^{-1} = g_0^{-1} - V_{\text{imp}}^\alpha. \quad (3)$$

The two leads are stacked block-diagonally, $g_{\text{rj}} = \text{diag}(g_{rL}, g_{rR})$, and coupled by the (transparency $\mathcal{T}$) tunnelling vertex

$$\Sigma_{\text{rj}} = \begin{pmatrix} 0 & \Sigma_{lr} \\ \Sigma_{rl} & 0 \end{pmatrix}, \quad \Sigma_{lr} = \Sigma_{rl}^\dagger \implies \Sigma_{\text{rj}}^\dagger = \Sigma_{\text{rj}}. \quad (4)$$

The full retarded GF is the Dyson solution

$$G^R \equiv G_{\text{rjwar}} = (\mathbb{I} - g_{\text{rj}} \Sigma_{\text{rj}})^{-1} g_{\text{rj}} = (g_{\text{rj}}^{-1} - \Sigma_{\text{rj}})^{-1}, \quad G^{R-1} = g_{\text{rj}}^{-1} - \Sigma_{\text{rj}}. \quad (5)$$

2 The equilibrium requirement

In equilibrium the Keldysh (Dyson) form and the fluctuation–dissipation form of the lesser GF must coincide,

$$G^< = G^R \Sigma^< G^A \quad \text{and} \quad G^< = f(G^A - G^R), \quad (6)$$

with $f(\omega) = \theta(-\omega) = ff$ at $T = 0$. The two agree *iff* the lesser self-energy is the FDT partner of the retarded one,

$$\Sigma^< = f(\Sigma^A - \Sigma^R), \quad \Sigma^A - \Sigma^R = G^{R-1} - G^{A-1}, \quad (7)$$

the last equality because the bare (Hermitian) inverse propagator cancels in $\Sigma^A - \Sigma^R$. Equation (7) is the single condition a correct $\Sigma^<$ must satisfy.

3 What $ff(G^A - G^R)$ evaluates to

Start from the algebraic identity, valid for any invertible G ,

$$G^A - G^R = G^R(G^{R-1} - G^{A-1})G^A, \quad (8)$$

which one checks by expanding $G^R G^{R-1} G^A - G^R G^{A-1} G^A = G^A - G^R$. Multiplying by ff ,

$$ff(G^A - G^R) = G^R \underbrace{[ff(G^{R-1} - G^{A-1})]}_{\equiv \Sigma_A^<} G^A. \quad (9)$$

So the FDT form *is* the bath form, with effective self-energy $\Sigma_A^< = ff(G^{R-1} - G^{A-1})$. Now evaluate it using (5). Two cancellations occur.

(i) The tunnelling drops out. Since Σ_{rj} is Hermitian (4),

$$G^{R-1} - G^{A-1} = (g_{\text{rj}}^{-1} - \Sigma_{\text{rj}}) - (g_{\text{rj}}^{A-1} - \Sigma_{\text{rj}}) = g_{\text{rj}}^{-1} - g_{\text{rj}}^{A-1}. \quad (10)$$

(ii) The impurity drops out. Per lead, using (3) and $V_{\text{imp}}^\dagger = V_{\text{imp}}$,

$$g_{r\alpha}^{-1} - g_{a\alpha}^{-1} = (g_0^{-1} - V_{\text{imp}}^\alpha) - (g_0^{A-1} - V_{\text{imp}}^\alpha) = g_0^{-1} - g_0^{A-1}. \quad (11)$$

Both leads give the *same* block, so

$$\Sigma_A^< = ff \begin{pmatrix} g_0^{-1} - g_0^{A-1} & 0 \\ 0 & g_0^{-1} - g_0^{A-1} \end{pmatrix}. \quad (12)$$

4 The surface identity

The BCS surface GF obeys the recursion $g_0^{-1} = Q(z) - \zeta^2 \tau_z g_0 \tau_z$ with $Q(z) = Q(z)^\dagger$ Hermitian (equivalently $g_0^{-1} + \zeta^2 \tau_z g_0 \tau_z$ is Hermitian; verified numerically to machine precision, $\lesssim 10^{-15}$). Subtracting the advanced version, the Hermitian Q cancels and

$$\boxed{g_0^{-1} - g_0^{A-1} = \zeta^2 \tau_z (g_0^A - g_0) \tau_z} \quad (13)$$

i.e. the *entire* anti-Hermitian part of g_0^{-1} is the reservoir term; there is no separate flat $-i\Gamma$ piece. Writing the clean spectral weight $g\ell_0 \equiv -(g_0 - g_0^A) = g_0^A - g_0$ and inserting (13) into (12),

$$\boxed{\Sigma_A^< = ff \zeta^2 \tau_z g\ell_0 \tau_z \quad \text{per lead}, \quad g\ell_0 = -(g_0 - g_0^A), \quad g_0 = g_0(\omega + i\Gamma).} \quad (14)$$

5 Comparison with the explicit Siglj

Equations (9) and (14) say that $ff(G^{R\dagger} - G^R)$ is exactly $G^R \Sigma_A^< G^A$ with the clean reservoir self-energy (14). Contrast with the self-energies that were tried in the code:

$\Sigma^<$ used	spectral in $\zeta^2 \tau_z (\dots) \tau_z$	lead dependence	flat $2i\Gamma$?
FDT / $\Sigma_A^<$ (correct)	bare $g_0(i\Gamma)$	none	absent
original (buggy)	<i>dressed</i> $g_{r\alpha} - g_{a\alpha}$	lead-dependent ($V_{\text{imp}}^L \neq V_{\text{imp}}^R$)	present
“clean + $2i\Gamma$ ”	bare $g_0(i\Gamma)$	none	present
“scheme B”	bare $g_0(i0^+)$	none	present

The correct $\Sigma_A^<$ therefore differs from the original code in two ways, both forced by the algebra above:

- **no V_{imp}** : the impurity cancels in $G^{R-1} - G^{A-1}$, so it must not enter $\Sigma^<$. Using the impurity-*dressed* spectral made $\Sigma^<$ lead-dependent ($V_{\text{imp}}^L \neq V_{\text{imp}}^R$), which broke the antisymmetry $I(-\phi) = -I(\phi)$ and produced a spurious static diode ($\eta \sim -2.4\%$) and a $\sim 1/\Gamma^2$ -divergent, $\sim 16\times$ inflated YSR supercurrent;
- **no $2i\Gamma$** : by (13) the whole broadening is already the reservoir term, so a separate flat $2i\Gamma$ has no retarded partner — it is a double count.

Remark (scheme A vs. B). Keeping Γ *inside* $g_0(\omega + i\Gamma)$ (“scheme A”, the boxed result) gives $\eta = 0$ and reproduces Ambegaokar–Baratoff for a clean junction, $I_c e R_N / \Delta \rightarrow \pi/2 \approx 1.57$. Moving Γ out of g_0 (evaluating the reservoir at $i0^+$) and re-adding it as a separate intrinsic broadening $-i\Gamma$ on the contact — with the matching retarded term $G^{R-1} = g_0(i0^+)^{-1} + i\Gamma - V_{\text{imp}}$ — is also self-consistent ($\eta = 0$) but over-broadens the gap edge and *fails* Ambegaokar–Baratoff ($I_c e R_N / \Delta \approx 0.60$); it is therefore not used.

Bottom line.

$\text{Gljwar} = \text{ff} .* (\text{Grjwar}' - \text{Grjwar}) \equiv \text{Grjwar} * \text{Siglj} * \text{Grjwar}'$ with $\Sigma_A^< = ff \zeta^2 \tau_z [-(g_0 - g_0^\dagger)] \tau_z$, g_0

The FDT form simply computes this clean, impurity-free, $2i\Gamma$ -free reservoir self-energy without building it by hand.